

Conclusion: Prospectively-collected swallowing metrics correlated with increased rates of clinically significant dysphagia in pts with T3-4 disease, or concurrent chemotherapy on the clinician-scored FOIS metric. However, post-treatment increase in dysphagia on the patient-reported EAT-10 score was associated with higher mean pharyngeal doses. Ongoing studies aim to further characterize treatment related parameters associated with the development of dysphagia.

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Bone Mineral Density Loss in Cervical Vertebrae Following Radiation Therapy for Head and Neck Cancers



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Purpose/Objective(s): Radiation therapy (RT) for the treatment of abdominal malignancy is associated with a significant reduction in bone mineral density (BMD) and vertebral compression fractures (VCFs) in thoracic and lumbar vertebrae. Since the total dose of radiation is greater for head and neck cancers, we sought to determine whether a similar relationship exists between RT, BMD, and VCFs in the vertebral bodies of the cervical spine in patients receiving RT for cancers of the head and neck.

Materials/Methods: A retrospective search identified 424 patients with cancers of the head or neck who underwent RT at our institution between May 1998 and August 2016. 105 of these patients received both a baseline CT scan and at least one CT scan after completing RT and were included in our analysis. Hounsfield units (HUs), which have been shown to be reliable approximations of dual X-ray absorptiometry results, were measured at each vertebral level from C1-T2 using sagittal CT images with contrast. Maximum radiation point dose to each vertebrae was measured on treatment planning software. In order to develop a more radiobiologically accurate dose-response model, all physical doses were converted to biologically equivalent dose using alpha/beta of 3 Gy. The percent change in HU ($\Delta\%HU$) was calculated between baseline and post-RT scans. Pearson's linear correlation was then used to compare $\Delta\%HU$ against the radiation point dose.

Results: The baseline BMD of each vertebral body in the cervical spine was at least 50% higher than that of those in the lower thoracic and lumbar spine. RT caused significant reductions in cervical vertebral BMD as measured by HU. There was no $\Delta\%HU$ within the first 6 months after RT. However, the $\Delta\%HU$ was significantly correlated with the radiation point dose to the vertebral body ($R = 0.288$, $p < 0.001$) within 6-12 months following RT. The same relationship persisted in subsequent follow-up scans 18-24 months following RT ($R = 0.43$, $p < 0.001$). At 18-24 months post-RT, our regression model showed that vertebral point doses of 5 Gy, 10 Gy, 20 Gy, 30 Gy, and 40 Gy caused $\Delta\%HU$ of 13.8%, 16.5%, 22.0%, 27.5%, and 33.0%, respectively. Zero of the 105 patients examined were found to have a VCF in the cervical spine.

Conclusion: Patients with cancers of the head or neck who undergo RT experience significant BMD losses in the vertebral bodies of the cervical spine. However, they do not have an increased risk of VCFs. This is likely due to a variety of factors, as the vertebral bodies of the cervical spine are not subject to the same degree of weight bearing and biomechanical stress as those of the thoracic and lumbar spine, have more collateral circulation and thus resistant to microvascular damage from radiation, and have a higher baseline BMD. Nevertheless, a model for post-RT BMD loss may help guide physicians in generating risk mitigation strategies in carefully selected patients.

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Evaluation of Tolerance to Robotic Radiosurgery-Based Stereotactic Radiation Therapy of Primary Intraocular Uveal Melanoma



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Purpose/Objective(s): This study was undertaken to assess the effect and evaluate tolerance on CyberKnife-based stereotactic radiotherapy treatment (CK-based SRT) by rate of complications in patient's group suffering from primary intraocular uveal melanoma. Doses for patients with intraocular tumor undergoing CK-based SRT are often challenging due to the close proximity of optic apparatuses.

Materials/Methods: We evaluate clinical outcomes and the dose plans of 20 patients treated by 3 to 14 with median 5 fractions. The PTV volume ranged from 1.20 - 202.90 cm³ (median 4.76 cm³, mean 21.76 cm³ $\pm$ SD 51.44). The prescription dose was 20-50 Gy with median value 50 Gy prescribed to the 80% isodose. The eye's fixation system with the polymetric device was used to fix the eye position. The device was attached to the mask system in cranio-caudal orientation and placed in front of the diseased eye. The patient was looking at the black point placed at the arm of the eye's fixation system. CK-based SRT of intraocular tumors were evaluated by: rate of complication, PTV coverage and conformity index (CI), the maximum dose to the optic nerve, lens, chiasma and lacrimal gland. Rate of complications is strongly correlated with tumor location to avoid theme it's require keeping dose within tolerance limits in such OAR's: lacrimal gland and cornea, lens, optic nerve, extraocular muscles, retina (or posterior eyewall at all), posterior chamber. Median follow-up was 25 months (range 11–25).

Results: The treatment was well tolerated by all patients and the stable position of the eye was obtained. Due to tumor proximity to OARs, at least one structure must be sacrificed to achieve proper tumor coverage. Within minimum 11-month follow-up period no grade-4 sequelae (requiring enucleation) was discovered. Table 1 showed rate of complications. Average treatment parameters were: PTV coverage 79.77 – 99.17 % (median 95.67 %, mean 94.51 % $\pm$ SD 5.07), CI 1.05 – 1.38 (median 1.13, mean 1.14 $\pm$ SD 0.08). The maximum dose to OARs were for: the optic nerve 9.87 – 55.36 Gy (median 29.87 Gy, mean 31.30 Gy $\pm$ SD 13.71), lens 2.10 – 35.49 Gy (median 7.48 Gy, mean 10.23 Gy $\pm$ SD 8.61), chiasma 0.87 – 30.65 Gy (median 6.96 Gy, mean 8.69 Gy $\pm$ SD 7.50) and lacrimal gland 6.50 – 51.34 Gy (median 16.31 Gy, mean 22.11 Gy $\pm$ SD 14.72).

Conclusion: The results of this study demonstrated good clinical outcome for CK-based SRT of intraocular tumors. More than one year follow up reported that the salvage therapy was unneeded in all cases. CK-based SRT appeared to be a relatively safe technique in treating intraocular tumors within or near optic apparatuses. CK-based SRT provides excellent tumor coverage without delivering excessive dose to OARs and provided that the dose to the optic apparatuses were limited.

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Intraocular complications	Grade-1	Grade-2	Grade-3
Kearatoconjunctivitis	5	2	1
Cataract formation	1	3	3
Radiation-induced retinopathy	10	5	3
Choroidal vasculopathy	11	6	1
Optic neuropathy	3	3	2
Scleral necrosis	1	1	0
Extraocular muscle deteriorations	1	0	0
Neovascular glaucoma	3	3	1